

TW1.3 - Docker & Jenkins Freestyle Project Report

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Clas: CS B

Subject: DevOps Lab

Aim

To build and deploy an application using Docker and Jenkins Freestyle Pipeline.

Objectives

Understand containerization, create Docker images, configure Jenkins jobs and automate builds.

Theory

Docker is a containerization platform that packages applications with their dependencies into lightweight containers. Jenkins is an automation server used for Continuous Integration and Continuous Deployment (CI/CD). Combining Docker with Jenkins enables automated building, testing and deployment of applications.

Software Requirements

Docker Desktop, Jenkins, Git, Java, Maven, GitHub.

Docker Concepts

Explain Docker Image, Container, Dockerfile, Docker Hub, Volumes, Networks and Bind Mounts.

Jenkins Concepts

Explain Freestyle Jobs, Build Triggers, Workspaces, Plugins, Build Steps and Post-build Actions.

Procedure

Install Docker and Jenkins, create Dockerfile, build image, configure Jenkins job, connect GitHub repository, execute build, verify successful deployment.

FileEditSelectionViewGoRunTerminalHelpmy-flask-app

app.py x Dockerfile requirements.txt 127.0.0.1:5000

```
app.py > ...
1 from flask import Flask
2
3 app = Flask(__name__)
4
5 @app.route("/")
6 def hello_world():
7     return "Hello, Docker!"
8
9 if __name__ == "__main__":
10     app.run(host="0.0.0.0", port=5000)
```

OUTPUTDEBUG CONSOLETERMINALPORTSQUERY RESULTSAZURE

```
PS C:\Users\Adarsh Jha\my-flask-app> docker build -t my-flask-app .
View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/jccxixsqk8ixuw2141ddf4no5
PS C:\Users\Adarsh Jha\my-flask-app> docker images
INFO In Use
IMAGEIDDISK USAGECONTENT SIZEEXTRA
docker.n8n.io/n8nio/n8n:latestcda6bafc7bb42.47GB387MBU
jenkins/jenkins:itsf4f65e6cd140801MB290MBU
jenkins:2.60.3eeb4850eb65f1.08GB316MB
my-flask-app:latest28e477584dbb200MB48.8MB
n8nio/n8n:latestcda6bafc7bb42.47GB387MBU
PS C:\Users\Adarsh Jha\my-flask-app> docker run -p 5000:5000 my-flask-app
Traceback (most recent call last):
  File "/app/app.py", line 1, in <module>
    from flask import Flask
  File "/usr/local/lib/python3.9/site-packages/flask/__init__.py", line 7, in <module>
    from .app import Flask as Flask
  File "/usr/local/lib/python3.9/site-packages/flask/app.py", line 27, in <module>
    from . import cli
  File "/usr/local/lib/python3.9/site-packages/flask/cli.py", line 17, in <module>
    from .helpers import get_debug_flag
  File "/usr/local/lib/python3.9/site-packages/flask/helpers.py", line 14, in <module>
    from werkzeug.urls import url_quote
```

Ln 10, Col 39Spaces: 4UTF-8CRLFPython

docker desktopPERSONAL

SearchCtrl+K

Ask Gordon BETAContainersImagesVolumesKubernetesBuildsModelsMCP Toolkit BETA Docker Hub Docker Scout Extensions

ContainersGive feedback

Container CPU usage0.43% / 800% (8 CPUs available)Container memory usage433.8MB / 3.66GBShow charts

SearchOnly show running containers

	Name	Container ID	Image	Port(s)	CPU (%)	Last started	Actions
☒	jenkins	cfdd6c610bc2	jenkins/jenkins:its	8080:8080	0.42%	1 hour ago	  
☐	n8n	c7fe654ac7a7	n8nio/n8n	5678:5678	0%	3 days ago	  
☐	hopeful_euclid	1e1cfccc76ba	my-flask-app	5000:5000	0%	1 hour ago	  
☒	brave_johnson	4470fc179832	my-flask-app	5000:5000	0.01%	1 hour ago	  

Showing 4 items

Walkthroughs

Multi-container applications8 mins

Containerize your application3 mins

View more in the Learning center

Engine runningRAM: 1.55 GBCPU: 0.00%Disk: 6.14 GB used (limit 1006.85 GB)TerminalUpdate available

my-flask-app

```
1 from flask import Flask
2
3 app = Flask(__name__)
4
5 @app.route("/")
6 def hello_world():
7     return "Hello, Docker!"
8
9 if __name__ == "__main__":
10     app.run(host="0.0.0.0", port=5000)
```

PS C:\Users\Adarsh Jha\my-flask-app> cd path\to\my-flask-app

PS C:\Users\Adarsh Jha\my-flask-app> docker --version
Docker version 29.1.3, build f52814d

PS C:\Users\Adarsh Jha\my-flask-app> docker ps

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
cfddcc610bc2	jenkins/jenkins:its	"/usr/bin/tini -- /u."	2 days ago	Up 8 minutes	0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp

PS C:\Users\Adarsh Jha\my-flask-app> docker build -t my-flask-app .

[+] Building 13.5s (11/11) FINISHED docker:desktop-linux

- > [internal] load build definition from Dockerfile 0.1s
- > transferring dockerfile: 211B 0.0s
- > [internal] load metadata for docker.io/library/python:3.9-slim 3.5s
- > [auth] library/python:pull token for registry-1.docker.io 0.0s
- > [internal] load .dockerignore 0.0s
- > transferring context: 2B 0.0s
- > [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53 4.6s
- > resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53 0.0s
- > sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8 29.78MB / 29.78MB 2.6s
- > sha256:fc74430849022d13b0d44b08969a953f842f59c6e9d1a0c2c83d71 13.88MB / 13.88MB 1.9s
- > sha256:b3ec39b36ae8c03a3e09e54de4ec4aa08381dfed84a9daa075048c2 1.29MB / 1.29MB 1.2s

PS C:\Users\Adarsh Jha\my-flask-app> docker build -t my-flask-app .

-> unpacking to docker.io/library/my-flask-app:latest 0.3s

View build details: [docker-desktop://dashboard/build/desktop-linux/desktop-linux/5yirnut6duudu37m1xp77kdzt](https://dashboard/build/desktop-linux/desktop-linux/5yirnut6duudu37m1xp77kdzt)

PS C:\Users\Adarsh Jha\my-flask-app> docker run -p 5000:5000 my-flask-app

- * Serving Flask app 'app'
- * Debug mode: off

WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.

- * Running on all addresses (0.0.0.0)
- * Running on http://127.0.0.1:5000
- * Running on http://172.17.0.3:5000

Press CTRL+C to quit

```
172.17.0.1 - - [27/Jul/2026 07:26:07] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [27/Jul/2026 07:26:07] "GET /favicon.ico HTTP/1.1" 404 -
172.17.0.1 - - [27/Jul/2026 07:28:15] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [27/Jul/2026 07:28:15] "GET /favicon.ico HTTP/1.1" 404 -
```

niccolkunzmann/first_timer_scraper#28 needs reviews 0 0 3 Connect Ln 10, Col 39 Spaces: 4 UTF-8 CRLF Python Python 3.14 (64

 Jenkins / All / New Item

Search

Help

Logout

New Item

Enter an item name

Flask-HelloWorld-Freestyle

Select an item type

 Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.

 Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

 Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

 Folder
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.

 Multibranch Pipeline
Creates a set of Pipeline projects according to detected branches in one SCM repository.

OK

 Jenkins / Flask-HelloWorld-Freestyle / #2

Search

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Logout

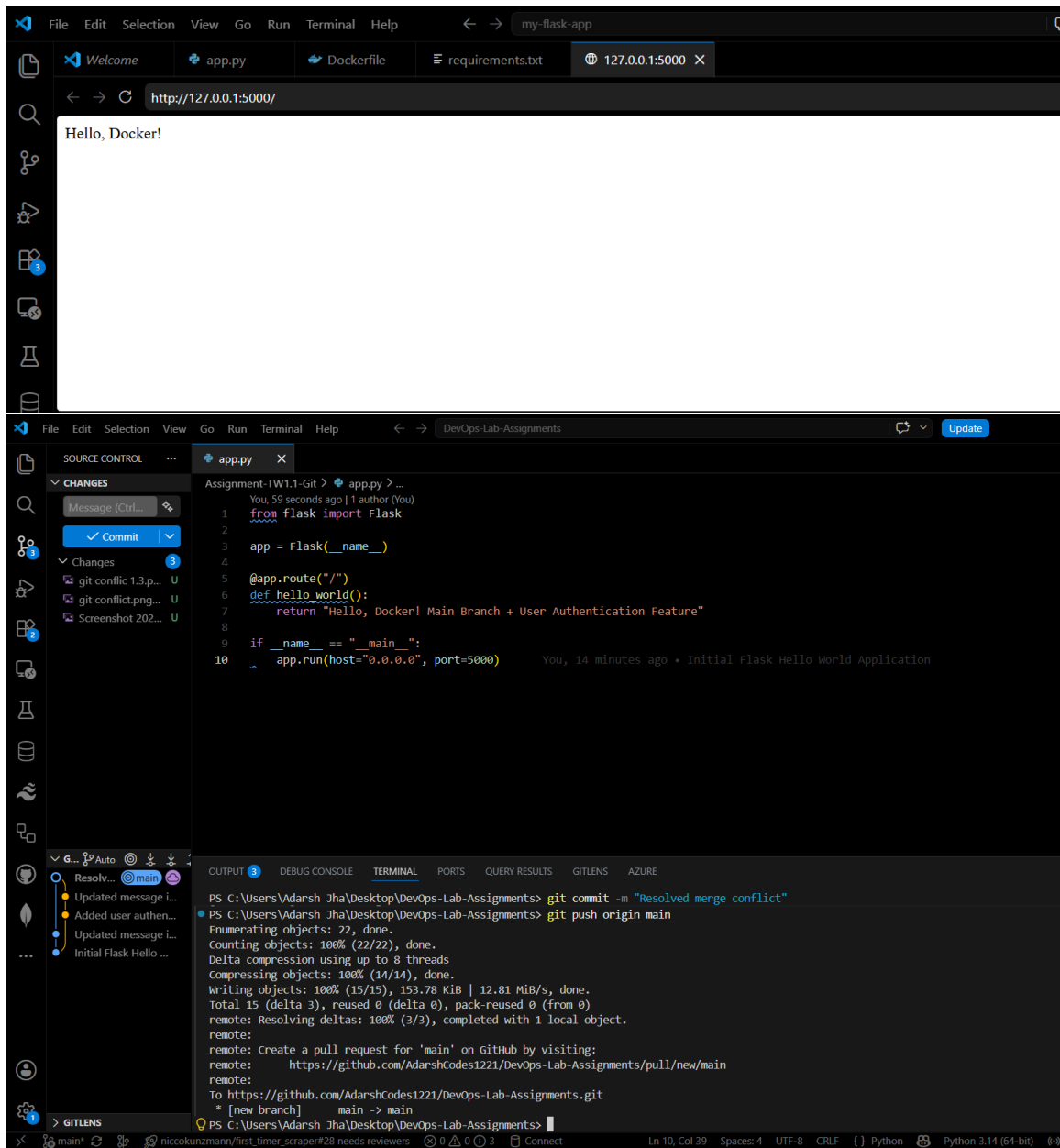
Console Output

Delete build '#2'

Git Build Data

Edit build information

```
Started by user Adarsh Jha
Running as SYSTEM
Building in workspace /var/jenkins_home/workspace/Flask-HelloWorld-Freestyle
The recommended git tool is: NONE
No credentials specified
> git rev-parse --resolve-git-dir /var/jenkins_home/workspace/Flask-HelloWorld-Freestyle/.git # timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url https://github.com/AdarshCodes1221/DevOps-Lab-Assignments.git # timeout=10
Fetching upstream changes from https://github.com/AdarshCodes1221/DevOps-Lab-Assignments.git
> git --version # timeout=10
> git --version # 'git version 2.47.3'
> git fetch --tags --force --progress -- https://github.com/AdarshCodes1221/DevOps-Lab-Assignments.git +refs/heads/*:refs/remotes/origin/* #
timeout=10
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 65f02dd9be6fe520b43d80a1b2254f553b843c5 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f 65f02dd9be6fe520b43d80a1b2254f553b843c5 # timeout=10
Commit message: "Resolved merge conflict"
First time build. Skipping changelog.
[Flask-helloworld-freestyle] $ /bin/sh -xe /tmp/jenkins13643275525767261008.sh
+ echo Listing workspace files
Listing workspace files
+ ls -la
total 20
drwxr-xr-x 5 jenkins jenkins 4096 Jul 27 08:51 .
drwxr-xr-x 8 jenkins jenkins 4096 Jul 27 08:50 ..
drwxr-xr-x 8 jenkins jenkins 4096 Jul 27 08:51 .git
drwxr-xr-x 2 jenkins jenkins 4096 Jul 27 08:51 Assignment-TW1.1-Git
drwxr-xr-x 2 jenkins jenkins 4096 Jul 27 08:51 Screenshots
Finished: SUCCESS
```



Commands Used

docker build, docker run, docker ps, docker images, docker stop, docker rm, docker logs, git clone, mvn clean package.

Learning Outcomes

Students learned Docker fundamentals, Jenkins automation and CI/CD workflow.

Conclusion

Docker and Jenkins together automate software delivery, improve consistency and reduce manual effort.